MIL-C-62/6B
19 December 1968
SUPERSEDING
62/6A
25 November 1964

MILITARY SPECIFICATION SHEET

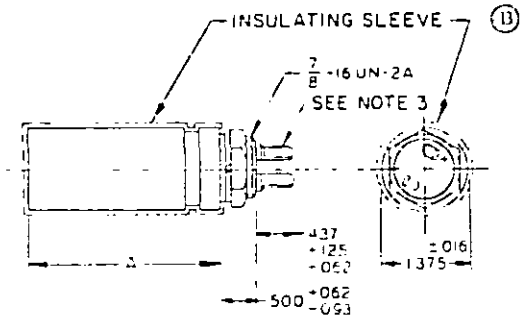
CAPACITORS, FIXED, ELECTROLYTIC

(DC, ALUMINUM, DRY ELECTROLYTE, POLARIZED)

STYLES CE41 AND CE44

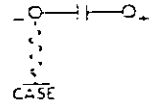
The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-62.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.



INCHES	MM
.016	.41
.062	1.57
.093	2.36
.125	3.18
.437	11.10
.500	12.70
1.375	34.93

STYLES
CE41 – uninsulated
CE44 – insulated



UNINSULATED STYLE NOT FOR NEW DESIGN
FOR REPLACEMENT PURPOSES ONLY

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Shape and position of lugs optional.
4. Mounting nut and lockwasher shall be supplied with each capacitor (Family 3)
5. There is an indeterminate resistance between the metal case and the negative terminal.

(13) denotes changes

FSC 5910

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Ⓑ TABLE I. Styles CE41 and CE44 capacitors.

Type designation 1/	Capacitance	DC rated voltage	DC leakage	Maximum impedance at 120 $^{+10}_{-5}$ Hz at -40° $^{+0°}_{-3°}$ C	Dissipation factor	Case dimension 2/ A 3/ ±.062 (1.57)
	uf	volts	ua	ohms	percent	inches
CE4-C101E	100	15	58	280	20	1.750 (44.45)
CE4-C271E	270	15	95	104	20	1.750 (44.45)
CE4-C471E	470	15	126	60	20	1.750 (44.45)
CE4-C681E	680	15	153	41	20	2.250 (57.15)
CE4-C102E	1000	15	183	28	20	2.250 (57.15)
CE4-C152E	1500	15	222	19	20	2.250 (57.15)
CE4-C222E	2200	15	273	13	20	2.250 (57.15)
CE4-C332E	3300	15	332	9.0	20	2.250 (57.15)
CE4-C472E	4700	15	399	6.0	20	3.250 (82.55)
CE4-C682E	6800	15	480	4.1	20	4.250 (107.95)
CE4-C270F	27	25	39	550	20	1.750 (44.45)
CE4-C470F	47	25	51	314	20	1.750 (44.45)
CE4-C680F	68	25	62	218	20	1.750 (44.45)
CE4-C101F	100	25	75	148	20	1.750 (44.45)
CE4-C271F	270	25	123	55	20	1.750 (44.45)
CE4-C471F	470	25	162	31	20	2.250 (57.15)
CE4-C681F	680	25	195	22	20	2.250 (57.15)
CE4-C911F	910	25	226	16	20	2.250 (57.15)
CE4-C102F	1000	25	237	14.8	20	2.250 (57.15)
CE4-C122F	1200	25	260	12.3	20	2.750 (69.85)
CE4-C152F	1500	25	291	9.9	20	3.000 (76.20)
CE4-C222F	2200	25	352	6.7	20	3.500 (88.90)
CE4-C332F	3300	25	423	4.5	20	4.250 (107.95)

See footnotes at end of table.

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Ⓑ TABLE I. Styles CE41 and CE44 capacitors. (Continued)

Type designation 1/	Capacitance	DC rated voltage	DC leakage	Maximum impedance at 120 $^{+10}_{-5}$ Hz at -40° $^{+0°}_{-3°}$ C	Dissipation factor	Case dimension 2/ A 3/ ±.062 (1.57)
	uf	volts	ua	ohms	percent	inches
CE4-C101G	100	50	106	148	15	1.750 (44.45)
CE4-C151G	150	50	130	99	15	1.750 (44.45)
CE4-C221G	220	50	157	67	15	2.250 (57.15)
CE4-C361G	360	50	201	41	15	2.250 (57.15)
CE4-C471G	470	50	229	31	15	2.250 (57.15)
CE4-C681G	680	50	276	22	15	2.250 (57.15)
CE4-C751G	750	50	290	20	15	2.250 (57.15)
CE4-C102G	1000	50	336	14.8	15	3.000 (76.20)
CE4-C152G	1500	50	411	9.9	15	3.500 (88.90)
CE4-C222G	2200	50	500	6.7	15	4.250 (107.95)
CE4-C680H	68	100	124	129	15	1.750 (44.45)
CE4-C101H	100	100	150	88	15	2.250 (57.15)
CE4-C151H	150	100	183	59	15	2.250 (57.15)
CE4-C221H	220	100	222	40	15	2.250 (57.15)
CE4-C271H	270	100	248	33	15	2.250 (57.15)
CE4-C331H	330	100	271	27	15	2.250 (57.15)
CE4-C431H	430	100	310	21	15	2.750 (69.85)
CE4-C471H	470	100	325	19	15	3.000 (76.20)
CE4-C681H	680	100	391	13	15	3.750 (95.25)
CE4-C470J	47	150	126	187	12	1.750 (44.45)
CE4-C331J	330	150	332	27	12	3.000 (76.20)
CE4-C471J	470	150	398	19	12	3.750 (95.25)

See footnotes at end of table.

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(B) TABLE I. Styles CE41 and CE44 capacitors. (Continued)

Type designation 1/	Capacitance	DC rated voltage	DC leakage	Maximum impedance at 120 $^{+10}_{-5}$ Hz at -40° $^{+0°}_{-3°}$C	Dissipation factor	Case dimension 2/ A 3/ ±.062 (1.57)
	uf	volts	ua	ohms	percent	inches
CE4-C680K	68	200	175	129	12	2.250 (57.15)
CE4-C820K	82	200	192	107	12	2.250 (57.15)
CE4-C101K	100	200	211	88	12	2.250 (57.15)
CE4-C151K	150	200	260	59	12	2.250 (57.15)
CE4-C221K	220	200	312	40	12	2.750 (69.85)
CE4-C331K	330	200	386	27	12	3.250 (82.55)
CE4-C471K	470	200	460	19	12	4.250 (107.95)
CE4-C101M	100	250	237	88	12	2.250 (57.15)
CE4-C151M	150	250	290	59	12	2.750 (69.85)
CE4-C221M	220	250	352	40	12	3.250 (82.55)
CE4-C331M	330	250	430	27	12	4.250 (107.95)
CE4-C750N	75	300	225	117	12	2.250 (57.15)
CE4-C121N	120	300	285	74	12	2.500 (63.50)
CE4-C151N	150	300	318	59	12	3.000 (76.20)
CE4-C221N	220	300	385	40	12	3.750 (95.25)
CE4-C100P	10	350	88	880	12	1.750 (44.45)
CE4-C270P	27	350	145	326	12	2.250 (57.15)
CE4-C470P	47	350	192	187	12	2.250 (57.15)
CE4-C680P	68	350	232	129	12	2.250 (57.15)
CE4-C820P	82	350	254	107	12	2.250 (57.15)
CE4-C101P	100	350	280	88	12	2.250 (57.15)
CE4-C121P	120	350	308	74	12	3.000 (76.20)
CE4-C151P	150	350	345	59	12	3.500 (88.90)

See footnotes at end of table.

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(B) TABLE I. Styles CE41 and CE44 capacitors. (Continued)

Type designation 1/	Capacitance	DC rated voltage	DC leakage	Maximum impedance at 120 $^{+10}_{-5}$ Hz at -40° $^{+0°}_{-3°}$ C	Dissipation factor	Case dimension 2/ A 3/ ±.062 (1.57)
	uf	volts	ua	ohms	percent	inches
CE4-C100Q	10	400	94	1480	17	1.750 (44.45)
CE4-C270Q	27	400	156	550	17	2.250 (57.15)
CE4-C330Q	33	400	172	450	17	2.250 (57.15)
CE4-C470Q	47	400	206	314	17	2.250 (57.15)
CE4-C680Q	68	400	248	218	17	2.500 (63.50)
CE4-C820Q	82	400	271	180	17	2.750 (69.85)
CE4-C101Q	100	400	300	148	17	3.250 (82.55)
CE4-C151Q	150	400	368	99	17	4.250 (107.95)
CE4-C100R	10	450	100	2200	17	2.250 (57.15)
CE4-C150R	15	450	123	1465	17	2.250 (57.15)
CE4-C270R	27	450	168	815	17	2.250 (57.15)
CE4-C330R	33	450	182	666	17	2.250 (57.15)
CE4-C470R	47	450	217	468	17	2.750 (69.85)
CE4-C680R	68	450	262	323	17	3.250 (82.55)
CE4-C820R	82	450	289	268	17	3.500 (88.90)
CE4-C101R	100	450	318	220	17	4.250 (107.95)

1/ Complete type designation shall include an additional digit to complete style.

2/ Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

3/ The dimensions shown are for style CE41; for style CE44, add .062 (1.57) ± .062 (1.57) to the nominal dimension.

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REQUIREMENTS:

Dimensions and configuration:
See figure and table I.

Case type: Tubular, uninsulated (CE41) and insulated (CE44).

Capacitance value: See table I.

(B) Capacitance tolerance:
At 200 volts, dc rated, and above:
-10, +50 percent.
At 150 volts, dc rated, and less:
-10, +75 percent.

Operating temperature range:
-40°C to +85°C.

(B) Solderability: Method 208 of MIL-STD-202.

DC leakage: See table I.

Capacitance: Method 305 of MIL-STD-202.

Dissipation factor: See table I.

Terminal strength: Method 211 of MIL-STD-202, test conditions A, B, and E.

Shock, medium impact: Method 205, test condition C, MIL-STD-202.

Vibration, high frequency: Method 204, test condition B, MIL-STD-202.

Salt spray (corrosion): Method 101, test condition B, MIL-STD-202.

Temperature cycling and immersion:
Temperature cycling: Method 102, test condition D, MIL-STD-202.
Immersion: Method 104, test condition C, MIL-STD-202.
DWV-Insulating sleeves-Method 301 of MIL-STD-202, 4,000 volts dc applied.
DCL - Not more than 150 percent of initial measurement.
Capacitance change - 5 percent.
DF - See table I.
Corrosion - Not more than 10 percent.

Moisture resistance: Method 106 of MIL-STD-202.
Insulation resistance (insulating sleeve) - Not less than 100 megohms.
DWV-Insulating sleeves-Method 301 of MIL-STD-202, 4,000 volts dc applied.
DCL - Not more than 150 percent of initial requirement.
Capacitance change - 5 percent.
DF - See table I.
Corrosion - Not more than 10 percent.

Stability at low and high temperature:
In accordance with MIL-C-62.
Step 1 (at +25°C):
DCL - See table I.
Capacitance - See table I.
DF - See table I.

Step 2 (at -40°C):
Impedance - See table I.

Step 3 (at +25°C):
DCL - See table I.
Capacitance - ±5 percent of initial value.
DF - See table I.

Step 4 (at +85°C):
DCL - Not more than 10 times initial value, or meet initial 25°C requirement.
Capacitance - ±20 percent of initial value.
DF - See table I.

Step 5 (at +25°C):
DCL - See table I.
Capacitance - ±5 percent of initial value.
DF - See table I.

Surge voltage: In accordance with MIL-C-62.
DCL - See table I.
Capacitance change - 6 percent.
DF - See table I.

Vent: Not applicable.

Life: Method 108, test condition F, MIL-STD-202.
DCL - See table I.
Capacitance change:
100 volts, dc rated, and less:
-20, +15 percent.
DC rated voltage above 100 volts:
±15 percent.
DF - Meet the initial requirement or not exceed 150 percent of the initial measurement.

Barometric pressure: Method 105 of MIL-STD-202.

Shelf life: In accordance with MIL-C-62.
Build factor: Not more than 4.0.
Shelf factor: Not more than 3.0.
DCL: Not more than 150 percent of initial value.
Capacitance change: 6 percent.
DF: Not more than 175 percent of initial requirement.

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Ⓑ TABLE II. Supersession data.

Superseded insulated part number	Superseded uninsulated part number	Recommended replacement style CE41 or CE44	Superseded insulated part number	Superseded uninsulated part number	Recommended replacement style CE41 or CE44
001	140	CE4-C101E	046	185	CE4-C331H
002	141	CE4-C271E	047	186	CE4-C431H
003	142	CE4-C471E	048	187	CE4-C471H
004	143	CE4-C681E	049	188	CE4-C681H
005	144	CE4-C102E	050	189	CE4-C100P
006	145	CE4-C152E	051	190	CE4-C270P
007	146	CE4-C152E	052	191	CE4-C330Q
008	147	CE4-C222E	053	192	CE4-C470J
009	148	CE4-C332E	054	193	CE4-C680K
010	149	CE4-C472E	055	194	CE4-C101K
011	150	CE4-C682E	056	195	CE4-C151K
012	151	CE4-C270F	057	196	CE4-C221K
013	152	CE4-C470F	058	197	CE4-C331J
014	153	CE4-C680F	059	198	CE4-C471J
015	154	CE4-C101F	060	199	CE4-C100P
016	155	CE4-C271F	061	200	CE4-C270P
017	156	CE4-C471F	062	201	CE4-C470P
018	157	CE4-C681F	063	202	CE4-C680K
019	158	CE4-C911F	064	203	CE4-C820K
020	159	CE4-C102F	065	204	CE4-C101K
021	160	CE4-C122F	066	205	CE4-C151K
022	161	CE4-C152F	067	206	CE4-C151K
023	162	CE4-C222F	068	207	CE4-C221K
024	163	CE4-C332F	069	208	CE4-C331K
025	164	CE4-C270P	070	209	CE4-C471K
026	165	CE4-C470J	071	210	CE4-C100P
027	166	CE4-C680H	072	211	CE4-C270P
028	167	CE4-C101G	073	212	CE4-C470P
029	168	CE4-C151G	074	213	CE4-C680P
030	169	CE4-C221G	075	214	CE4-C680P
031	170	CE4-C331H	076	215	CE4-C820P
032	171	CE4-C471G	077	216	CE4-C101M
033	172	CE4-C681G	078	217	CE4-C151M
034	173	CE4-C751G	079	218	CE4-C221M
035	174	CE4-C102G	080	219	CE4-C331M
036	175	CE4-C152G	081	220	CE4-C100P
037	176	CE4-C222G	082	221	CE4-C270P
038	177	CE4-C270P	083	222	CE4-C330Q
039	178	CE4-C470J	084	223	CE4-C470P
040	179	CE4-C680H	085	224	CE4-C680P
041	180	CE4-C101H	086	225	CE4-C750N
042	181	CE4-C151H	087	226	CE4-C101P
043	182	CE4-C151H	088	227	CE4-C121N
044	183	CE4-C221H	089	228	CE4-C151N
045	184	CE4-C271H	090	229	CE4-C221N

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Ⓑ TABLE II. Supersession data. (Continued)

Superseded insulated part number	Superseded uninsulated part number	Recommended replacement style CE41 or CE44	Superseded insulated part number	Superseded uninsulated part number	Recommended replacement style CE41 or CE44
091	230	CE4-C100P	103	242	CE4-C680Q
092	231	CE4-C270P	104	243	CE4-C820Q
093	232	CE4-C470P	105	244	CE4-C101Q
094	233	CE4-C680P	106	245	CE4-C151Q
095	234	CE4-C820P	107	246	CE4-C100R
096	235	CE4-C101P	108	247	CE4-C150R
097	236	CE4-C121P	109	248	CE4-C270R
098	237	CE4-C151P	110	249	CE4-C330R
099	238	CE4-C100Q	111	250	CE4-C470R
100	239	CE4-C270Q	112	251	CE4-C680R
101	240	CE4-C330Q	113	252	CE4-C820R
102	241	CE4-C470Q	114	253	CE4-C101R

Application data:
All capacitors shall withstand an rms ripple current at 120 Hz and 85° C as determined from the formula:

$$I_r = k\sqrt{C}$$

Where:

I_r = rms ripple current in milliamps.
C = nominal capacitance in microfarads.
k = 50 for units rated at 15 to 350 volts dc. incl.
= 40 for units rated at 400 and 450 volts dc.

Ⓑ Supersession data:
See table II.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU, MI
Navy - EC, AS, OS
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - ME, AT, AV
Navy - MC
Air Force - 19

Preparing activity:
Army - EL

(Project 5910-0848-7)

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(See Instructions -- Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* — Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* — Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* — Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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